

xD-69 Hardware Description



This amazing logo was designed by someone called kilésengati on Discord :)

CPU: NEC uPD78312 – 8/16 bit embedded microcontroller

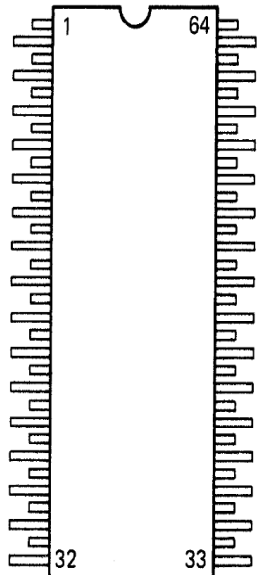
NEC 78k/3 series microcontroller for embedded devices. CISC(Complex instruction set computer) processor, little endian.

No current ROM dump. Has a PROM that contains opcode stuff I think

Serial MIDI transmission uses RXD and TXD CPU pins.

CPU

μPD78312

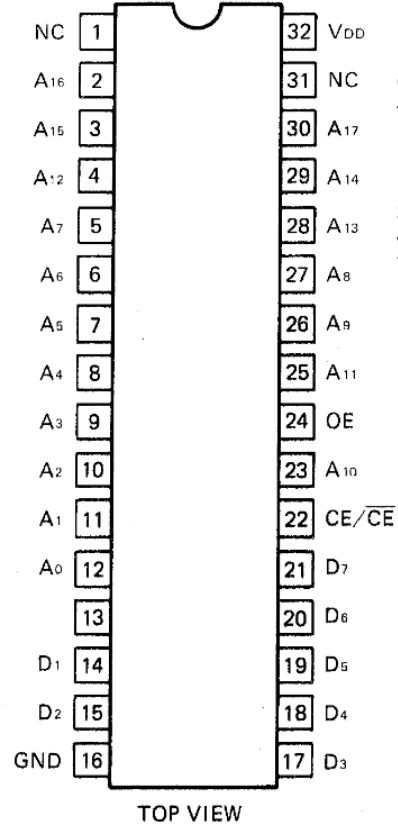


TOP VIEW

No	NAME	I/O	No	NAME	I/O	No	NAME	I/O	No	NAME	I/O
1	P0.0	O	17	NMI	I	33	AN0	I	49	A14	O
2	P0.1	O	18	INT0	I	34	AN1	I	50	A15	O
3	P0.2	O	19	INT1	I	35	AN2	I	51	$\overline{EA}$	I
4	P0.3	O	20	INT2	I	36	AN3	I	52	$\overline{RESET}$	I
5	P0.4	O	21	TXD	O	37	AVREF	O	53	$\overline{RD}$	O
6	P0.5	O	22	RXD	I	38	AVss	-	54	$\overline{WR}$	O
7	P0.6	O	23	$\overline{SCK}$	O (NC)	39	P3.4	I/O(NC)	55	ALE	O
8	P0.7	O	24	$\overline{CTS}$	I/O	40	P3.5	I/O(NC)	56	AD0	I/O
9	P1.0	I/O	25	$\overline{RFSH}$	O (NC)	41	P3.6	I/O	57	AD1	I/O
10	P1.1	I/O	26	P3.0	I	42	P3.7	I/O	58	AD2	I/O
11	P1.2	I/O	27	P3.1	I	43	A8	O	59	AD3	I/O
12	P1.3	I/O	28	P3.2	I	44	A9	O	60	AD4	I/O
13	P1.4	I/O	29	P3.3	I	45	A10	O	61	AD5	I/O
14	P1.5	I/O	30	X1	I	46	A11	O	62	AD6	I/O
15	P1.6	I/O	31	X2	I	47	A12	O	63	AD7	I/O
16	P1.7	I/O(NC)	32	Vss	-	48	A13	O	64	VDD	-

Read-Only Memories

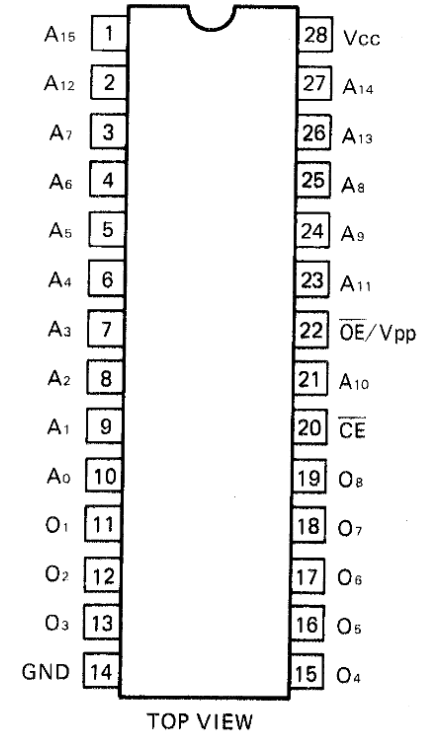
**PCM ROM A/B
TC532000**



PCM storage ROM – contains PCM partials for the preset playback. Has been ROM dumped. Roland owns the samples so you need to provide your own copy

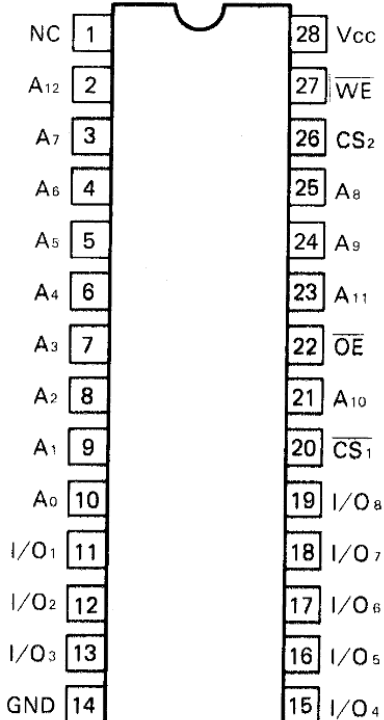
OS ROM - This is where the D-50 runs its code from for the D-50 operating system. I have provided the disassembly in another folder. I managed to disassemble the binary using MAME disasm command line

EP ROM MBM27C512



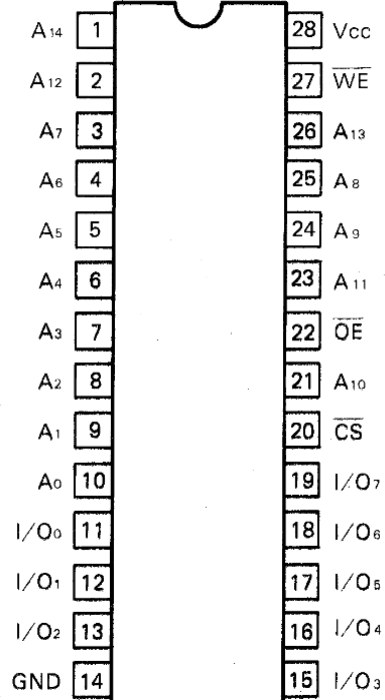
Random-Access Memories

**S RAM
HM6264ASP**



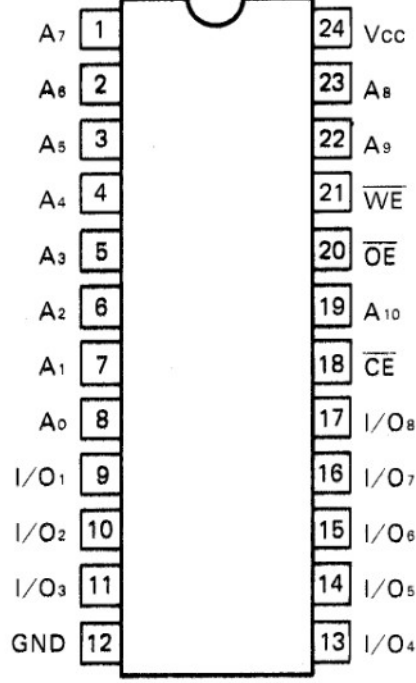
TOP VIEW

**S RAM
HM62256LP**



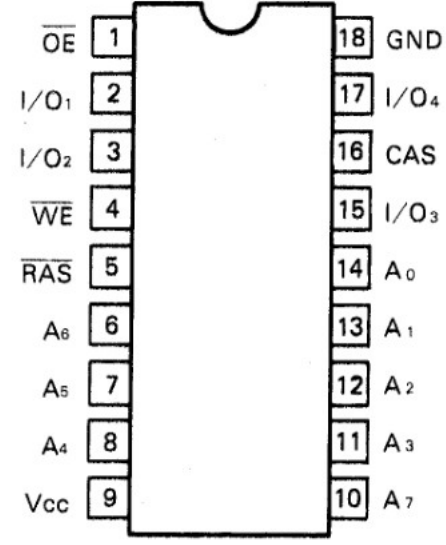
TOP VIEW

**S RAM
LC3517AS**



TOP VIEW

**D RAM
 μ PD41416**

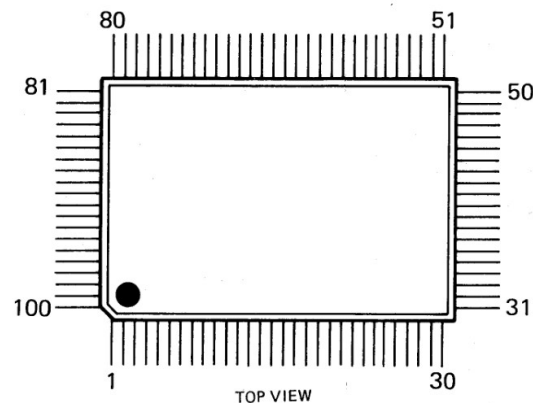


TOP VIEW

Target Custom ASIC – Chorus DSP

Please see:
<https://github.com/pxlsr/FujiCells>
for a detailed and comprehensive
documentation of the ASICs

**CHORUS CUSTOM IC
MB87137**

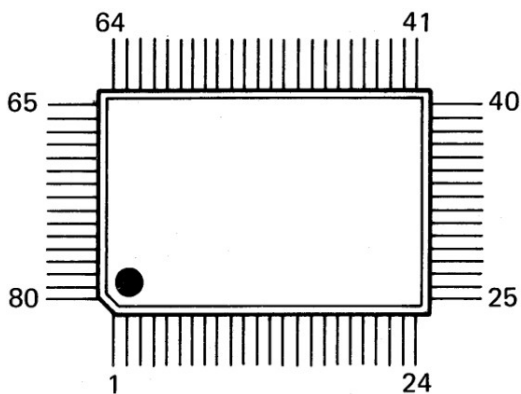


Pin No.	Pin Name	Input/Output	Description	Pin No.	Pin Name	Input/Output	Description
1	RES	I	Reset input	51	O8	O	Data input
2	E	I	Chip enable input	52	O9	O	Data input
3	VDD	-	+5v	53	VDD	-	+5v
4	CS	I	Chip Select input	54	O10	O	Data input
5	RW	I	Write Pulse input	55	O11	O	Data input
6	RD	I	Read Pulse input	56	O12	O	Data input
7	CS	I	Chip select input	57	O13	O	Data input
8	A0	I	Connect to CPU address bus	58	O14	O	Data input
9	A1	I	Connect to CPU address bus	59	O15	O	Data input
10	A2	I	Connect to CPU address bus	60	RA0	O	Connects to SRAM address bus RA13 not used
11	D0	I/O	Connect to CPU data bus	61	WE	O	SRAM write pulse output
12	D1	I/O	Connect to CPU data bus	62	RA1	O	Connects to SRAM address bus RA13 not used
13	D2	I/O	Connect to CPU data bus	63	RA2	O	Connects to SRAM address bus RA13 not used
14	D3	I/O	Connect to CPU data bus	64	RA3	O	Connects to SRAM address bus RA13 not used
15	VSS	-	GND	65	VSS	-	GND
16	D4	I/O	Connect to CPU data bus	66	RA4	O	Connects to SRAM address bus RA13 not used
17	D5	I/O	Connect to CPU data bus	67	RA5	O	Connects to SRAM address bus RA13 not used
18	D6	I/O	Connect to CPU data bus	68	RA6	O	Connects to SRAM address bus RA13 not used
19	D7	I/O	Connect to CPU data bus	69	RA7	O	Connects to SRAM address bus RA13 not used
20	DOE	I	Data out enable input	70	RA8	O	Connects to SRAM address bus RA13 not used
21	INCK	I	Input data latch clock input	71	OE	O	SRAM out enable output
22	SIN	I	Sync input	72	RA9	O	Connects to SRAM address bus RA13 not used
23	SOUT	O	Sync output	73	RA10	O	Connects to SRAM address bus RA13 not used
24	LRS	I	L/R select input	74	RA11	O	Connects to SRAM address bus RA13 not used
25	I0	I	Data input	75	CE	O	SRAM chip enable output
26	I1	I	Data input	76	RA12	O	Connects to SRAM address bus RA13 not used
27	I2	I	Data input	77	PD7	I/O	Connect to SRAM data bus
28	VDD	-	+5v	78	VDD	-	+5v
29	I3	I	Data input	79	RA13	O	Connects to SRAM address bus RA13 not used
30	I4	I	Data input	80	PD6	I/O	Connect to SRAM data bus
31	I5	I	Data input	81	PD5	I/O	Connect to SRAM data bus
32	I6	I	Data input	82	PD4	I/O	Connect to SRAM data bus
33	I7	I	Data input	83	PD3	I/O	Connect to SRAM data bus
34	I8	I	Data input	84	PD2	I/O	Connect to SRAM data bus
35	I9	I	Data input	85	PD1	I/O	Connect to SRAM data bus
36	I10	I	Data input	86	PD0	I/O	Connect to SRAM data bus
37	I11	I	Data input	87	VSS	-	GND
38	I12	I	Data input	88	X1	I	Master clock input
39	I13	I	Data input	89	X2	O	not used
40	VSS	-	GND	90	VSS	-	GND
41	I14	I	Data input	91	ROMT	I	No description
42	I15	I	Data input	92	RAMT	I	No description
43	O0	O	Data input	93	CTRT	I	No description
44	O1	O	Data input	94	THRU	I	No description
45	O2	O	Data input	95	ECTL	I	External control select input
46	O3	O	Data input	96	ADDA	I	No description
47	O4	O	Data input	97	OFST	I	offset binary select input
48	O5	O	Data input	98	PSFT	I	No description
49	O6	O	Data input	99	LHLD	O	Signal output for S/H – not used
50	O7	O	Data input	100	RHLD	O	Signal output for S/H – not used

Target Custom ASIC – Reverb DSP

Please see:
<https://github.com/pxlsr/FujiCells>
 for a detailed and comprehensive
 documentation of the ASICs

REVERB CUSTOM IC MB87126-006



TOP VIEW

PIN.NO.	PIN NAME	I/O	DESCRIPTION	PIN.NO.	PIN NAME	I/O	DESCRIPTION
1, 2, 66~72, 74~80	DC0-15	O	Data output for chorus chip and DAC D/A へのデータ, コーラス・データ出力端子	20	LOAD	O	Sync signal output シンク信号出力端子
3	STRT	I	Pulled low GND にプルダウン	21	SYNC	I	Sync signal input シンク信号入力端子
4	DIN	I	Pulled low GND にプルダウン	22	INCK	I	Data latch clock input for initialization イニシャライズ時のデータ・ラッチ・クロック入力端子
5	CLEA	I	Pulled low GND にプルダウン	23	ERCL	I	Busy veset output Busy 解除用端子
6~10	RD0-4	O	Control output for enable and for S/H and Lower for bit D/A Conversion コントロール出力端子 イネーブル, S/H, D/A (下4 bit)	24	BUSY	O	Serial data transfer error output (Parity check) シリアル・データ転送エラー出力 (パリティ・チェック)
11	RSET	I	Pulled low GND にプルダウン	25	SXD	I	Serial data input シリアル・データ入力端子
12, 15, 36, 52, 65	Vss	—	GND	26	SCK	I	Serial data read-in clock input シリアル・データ取込みクロック入力端子
13	SLRQ	I	Pulled low GND にプルダウン	27 - 32, 34, 35,	DAO-7	O	Connect to RAM address bus RAM アドレス・バス
14	MSCK	I	Master clock input マスター・クロック入力端子	37	RAS	O	Row address strobe output ロー・アドレス・ストロブ
16	SLCK	O	Not used 未使用	38	WE	O	DRAM write pulse output DRAM ライト・パルス出力端子
17	TEST	I	Pulled low GND にプルダウン	39	CAS	O	Column address strobe output カラム・アドレス・ストロブ
18	TMB	O	Time base signal output タイム・ベース信号出力端子	40 - 51, 53 - 64,	DRO-23	I/O	Connect to RAM data bus Synth and Chorus data input RAMデータ・バス, シンセ, コーラスデータ入力端子
19, 33, 73	VDD	—	+5 V				

Target Custom ASIC – Synth (DSP core)

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SYNTH CUSTOM IC MB87136

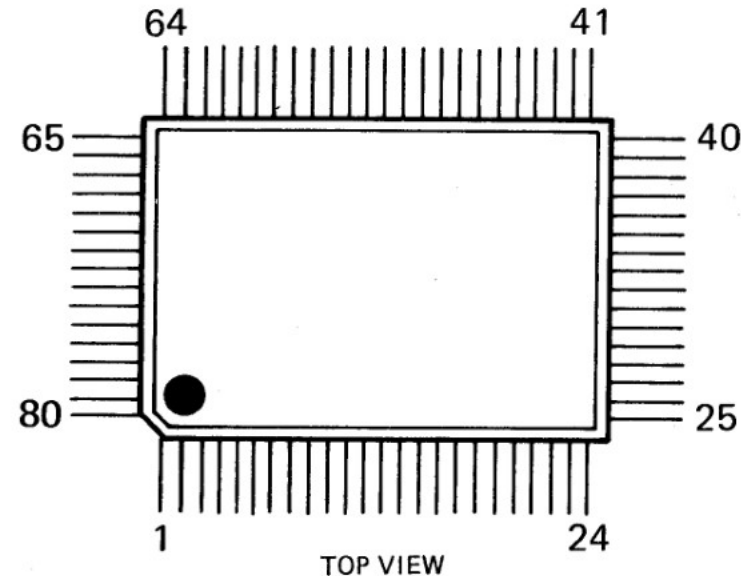
34	33	32	31	30	29	28	27	26	25	24	23
35	72	71	70	69	68	67	66	65	64	63	22
36	73				86	85				62	21
37	74									61	20
38	75									60	19
39	76	87							84	59	18
40	77	88							83	58	17
41	78									57	16
42	79									56	15
43	80				81	82				55	14
44	45	46	47	48	49	50	51	52	53	54	13
1	2	3	4	5	6	7	8	9	10	11	12

TOP VIEW

PIN.NO.	PIN NAME	I/O	DESCRIPTION	PIN.NO.	PIN NAME	I/O	DESCRIPTION
1	CS	I	Chip select チップ・セレクト入力端子	44	INT	O	Interrupt output インタラプト 出力端子
2 - 6, 46 - 49,	A0-8	I	Connect to CPU address bus CPU とのアドレス・バス	45	OE	I	Output enable input アウトプット・イネーブル入力端子
7 - 10, 50 - 53,	D0-7	I/O	Connect to CPU data bus CPU とのデータ・バス	75		-	Not used 未使用
11 - 14, 54 - 57,	PD0-7	I	Connect to ROM data bus ROM とのデータ・バス	76	X2	I/O	Xtal input 水晶振動子 (32.768 MHz) 接続端子
15 - 26, 58 - 65,	RA0-19	O	Connect to ROM address bus ROM とのアドレス・バス	77	16M	O	Output frequency is one half of master clock マスター・クロックを1回分周した周波数を出力
27 - 35, 66 - 72,	O0-15	O	Data output データ・アウトプット・バス	78	CKIN	I	lutput frequency is a combination of the master clock and one half of master clock マスター・クロックと1回分周した周波数を出力 Not used
36 - 37, 73 - 74,	SH0-3	O	Not used 未使用	79		-	Not used 未使用
38		-	Not used 未使用	80	RD	I	Read pulse input リード・パルス入力端子
39	X1	I/O	Xtal input (32.768 MHz) 水晶振動子 (32.768 MHz) 接続端子	81, 84, 85, 88,	Vss	-	GND
40	32M	O	The same frequency as that of master clock マスター・クロックと同じ周波数を出力	82, 83, 86, 87,	VDD	-	+5 V
41		-	Not used 未使用				
42	SYI	I	Sync signal input シンク信号入力端子				
43	WR	I	Write pulse input ライト・パルス入力端子				

Target Custom Gate Array – LCD controller with possible peripheral control logic?

**GATE ARRAY
HG61H25B18F**

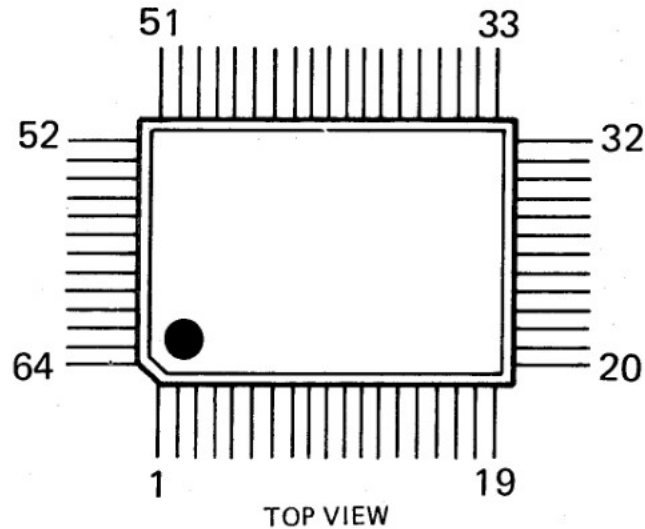


PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O
1	SYNT2	O (NC)	21	ALE	I	41	EC	O	61	R2	I
2	IRAM	O (NC)	22	$\overline{WR}$	I	42	O0	O	62	R3	I
3	RAM	O	23	$\overline{RD}$	I	43	O1	O	63	R4	I
4	A7	O	24	$\overline{RESET}$	I	44	O2	O	64	R5	I
5	A6	O	25	A15	I	45	O3	O	65	R6	I
6	A5	O	26	A14	I	46	O4	O	66	R7	I
7	A4	O	27	A13	I	47	O5	O	67	CORUS	O
8	A3	O	28	A12	I	48	O6	O	68	SCK	O
9	A2	O	29	A11	I	49	O7	O	69	SXD	O
10	A1	O	30	A10	I	50	S0	O	70	BUSY	I
11	A0	O	31	A9	I	51	S1	O	71	ERCL	O
12	Vss	-	32	A8	I	52	Vss	-	72	LOAD	I
13	AD7	I/O	33	VDD	-	53	S2	O	73	VDD	-
14	AD6	I/O	34	ARS	I (HIGH)	54	S3	O	74	TMB	I
15	AD5	I/O	35	INT1	O (NC)	55	S4	O	75	SINT1	I (LOW)
16	AD4	I/O	36	INT2	O	56	S5	O	76	SINT2	I (LOW)
17	AD3	I/O	37	DSCAN	O	57	S6	O	77	TEST1	I (LOW)
18	AD2	I/O	38	$\overline{ERAM}$	O	58	S7	O	78	CLK	I
19	AD1	I/O	39	ERAM	O (NC)	59	R0	I	79	TEST2	I (LOW)
20	AD0	I/O	40	RS	O	60	R1	I	80	SYNT1	O

Target Custom Gate Array – Memory card controller

GATE ARRAY

μ PD65005G-062

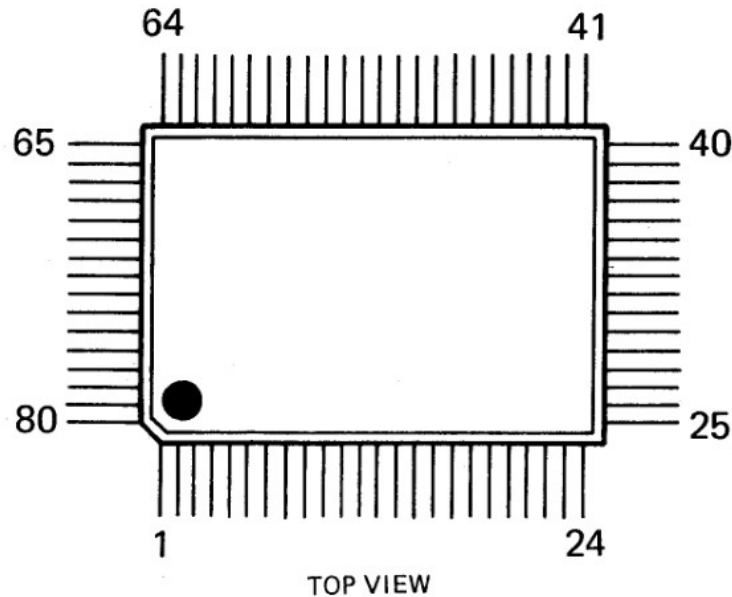


PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O
1	NC	-	17	NC	-	33	NC	-	49	NC	-
2	NC	-	18	NC	-	34	NC	-	50	CD0	I/O
3	AD7	I/O	19	A13	I	35	CA5	O	51	CD1	I/O
4	AD6	I/O	20	A12	I	36	CA6	O	52	CD2	I/O
5	AD5	I/O	21	A11	I	37	CA7	O	53	CD3	I/O
6	AD4	I/O	22	A10	I	38	CA8	O	54	CD4	I/O
7	AD3	I/O	23	A9	I	39	CA9	O	55	CD5	I/O
8	AD2	I/O	24	A8	I	40	CA10	O	56	CD6	I/O
9	AD1	I/O	25	SEL	I (LOW)	41	CA11	O	57	CD7	I/O
10	AD0	I/O	26	Vss	-	42	CA12	O	58	Vss	-
11	Vss	-	27	VDD	-	43	CA13	O	59	VDD	-
12	VDD	-	28	CA0	O	44	CA14	O	60	BATT	I (LOW)
13	ALE	I	29	CA1	O	45	MR	O	61	SENS	I (NC)
14	$\overline{\text{WR}}$	I	30	CA2	O	46	CWR	O	62	RCS	I
15	$\overline{\text{RD}}$	I	31	CA3	O	47	CCS	O	63	$\overline{\text{CS}}$	I
16	A14	I	32	CA4	O	48	CRD	O	64	NC	-

Target Custom Gate Array

– Keyboard controller

**GATE ARRAY
MB63H149**



PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O
1	T7	O	21	BR9	I	41	AD7	I/O	61	RA1	O
2	BR0	I	22	MK9	I	42	CA8	I	62	RA10	O
3	MK0	I	23	BR10	I	43	CA9	I	63	RA2	O
4	BR1	I	24	MK10	I	44	CA10	I (LOW)	64	$\overline{ROE}$	I/O
5	MK1	I	25	RES	I	45	CS	I	65	RA3	O
6	BR2	I	26	$\overline{EXCK}$	I/O	46	XT1	I	66	$\overline{RWE}$	O
7	MK2	I	27	E	I (HIGH)	47	XT2	O (NC)	67	RA4	O
8	BR3	I	28	INT	O	48	ASEL	O (NC)	68	RA9	O
9	MK3	I	29	AS	I	49	MOD1	I (HIGH)	69	RA5	O
10	BR4	I	30	$\overline{CRES}$	O (NC)	50	MOD2	I (LOW)	70	RA8	O
11	MK4	I	31	$\overline{CRNW}$	I	51	RD3	I/O	71	RA6	O
12	Vss	-	32	SRCK	O (NC)	52	Vss	-	72	RA7	O
13	BR5	I	33	VDD	-	53	RD4	I/O	73	VDD	-
14	MK5	I	34	AD0	I/O	54	RD2	I/O	74	T0	O
15	BR6	I	35	AD1	I/O	55	RD5	I/O	75	T1	O
16	MK6	I	36	AD2	I/O	56	RD1	I/O	76	T2	O
17	BR7	I	37	AD3	I/O	57	RD6	I/O	77	T3	O
18	MK7	I	38	AD4	I/O	58	RD0	I/O	78	T4	O
19	BR8	I	39	AD5	I/O	59	RD7	I/O	79	T5	O
20	MK8	I	40	AD6	I/O	60	RA0	O	80	T6	O